NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

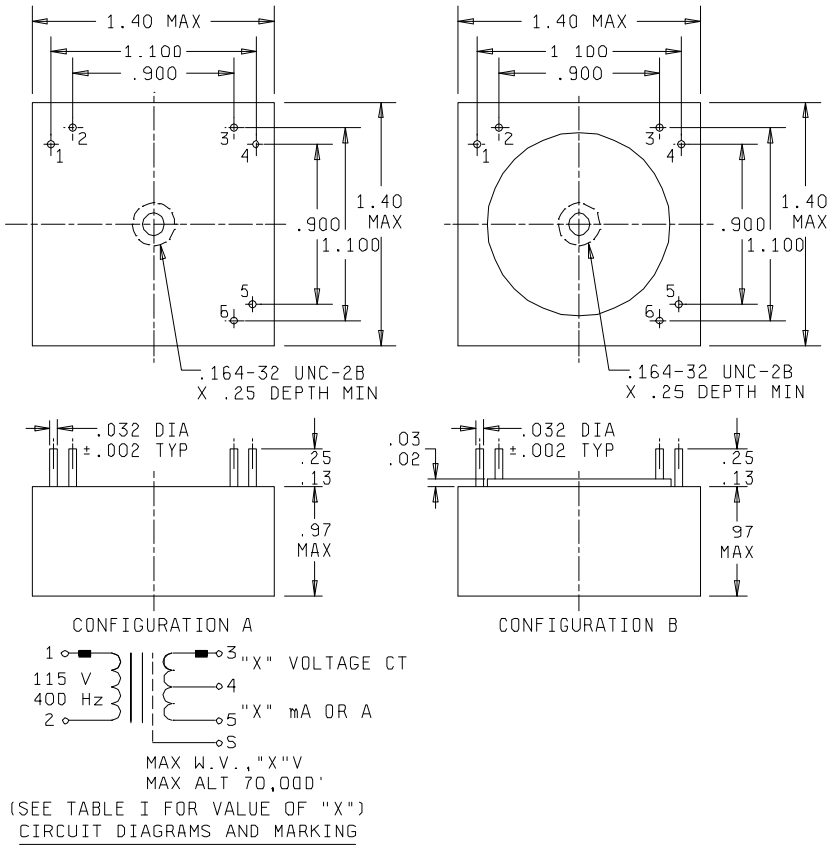
MIL-PRF-27/45B
7 February 1984
SUPERSEDING
MIL-T-27/45A
20 October 1976

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 16 VOLT AMPERES, 400 HZ,
ELECTROSTATIC SHIELDED, TF5S03ZZ752 TO TF5S03ZZ795, INCLUSIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the transformers described herein shall consist of this specification and the latest issue of MIL-T-27.



INCHES	MM
.002	0.05
.02	0.5
.03	0.8
.032	0.81
.13	3.3
.164	4.17
.25	6.4
.900	22.86
.97	24.6
1.00	25.4
1.100	27.94
1.40	35.6

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.005 (0.13 mm).
4. Marking shall be on the side of the case.
5. A white dot shall be marked on the side opposite the terminals, over terminal 1, in addition to the terminal marking shown.
6. Secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configuration.

(B) denotes changes

FSC 5950

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REQUIREMENTS: (When numbers in parentheses i.e. (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings. See table I.

Primary voltage (1-2): 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction:

Dimensions and configuration. See figure 1.

Weight. 3.4 ounces, maximum.

Duty cycle: Continuous.

Case. Plastic encapsulated.

Material: Plastic.

Altitude: 70,000 feet, maximum.

Operating temperature range: -55°C to +130°C maximum.

Terminals. Printed circuit pins - tinned brass.

Terminal strength. Method 211 of MIL-STD-202, test condition A, 2.0 pounds.

Dielectric withstanding voltage.

At sea level 2.8 times the maximum working voltage specified in table I.

At reduced barometric pressure. 1.25 times the maximum working voltage specified in table I.

Electrical characteristics.

No load With 115 volts, 400 Hz across (1-2): Current in (1-2) shall not exceed 16 mA rms.

Rated load. With 115 volts, 400 Hz across (1-2) and a resistive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I.

Efficiency 87 percent minimum at rated load.

Electrostatic shielding. The voltage ratio shall be a minimum of 5 to 1 at 20 kHz.

Polarity. Additive with terminals 2 and 3 connected.

Regulation. $\frac{\text{Voltage (no load)} - \text{Voltage (rated load)}}{\text{Voltage (rated load)}} \times 100$ shall not exceed 8 percent.

Phase shift: 1.0 electrical degree between primary and secondary voltage at rated load, maximum.

Temperature rise. 32°C, maximum.

Vibration: MIL-STD-202, method 204, test condition D.

Marking location. See figure 1.

Ⓑ QUALITY ASSURANCE PROVISIONS.

Extent of qualification.

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Qualification testing and approval to M27/37-44 A or B, M27/42-44 A or B, and M27/46-44 A or B shall be sufficient to grant qualification approval to MIL-T-27/37 through MIL-T-27/47 inclusive, all parts.

Qualification testing and approval to M27/45-44 shall be sufficient to grant qualification approval to M27/45-01 through M27/45-44.

Part number example:

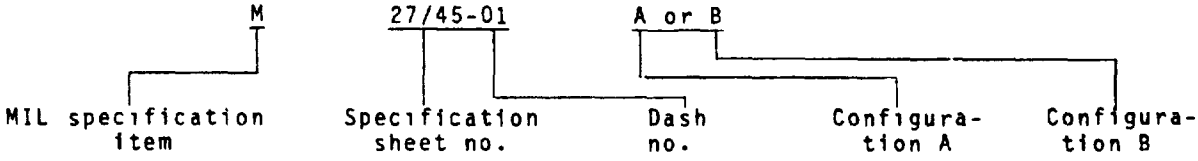


TABLE I. Electrical ratings.

Dash no.	Type designation	Secondary voltage (3-5) (V rms) ±3%	Secondary current (3-5) (amps rms)	Working voltage (V rms max)
01 A or B	752	6	2.667	325
02 "	753	6.3	2.540	"
03 "	754	7	2.286	"
04 "	755	7.5	2.133	"
05 "	756	8	2.000	"
06 "	757	9	1.778	"
07 "	758	10	1.600	"
08 "	759	11	1.455	"
09 "	760	12	1.333	"
10 "	761	12.6	1.270	"
11 "	762	13	1.231	"
12 "	763	15	1.067	"
13 "	764	16	1.000	"
14 "	765	18	0.889	"
15 "	766	20	0.800	"
16 "	767	22	0.727	"
17 "	768	24	0.667	"
18 "	769	26	0.615	"
19 "	770	28	0.571	"
20 "	771	30	0.533	"
21 "	772	33	0.485	"
22 "	773	36	0.444	"
23 "	774	39	0.410	"
24 "	775	43	0.372	"
25 "	776	47	0.340	"
26 "	777	51	0.314	"
27 "	778	56	0.286	"
28 "	779	62	0.258	"
29 "	780	68	0.236	"
30 "	781	75	0.213	"
31 "	782	82	0.195	"
32 "	783	91	0.176	"
33 "	784	100	0.160	"
34 "	785	110	0.145	"
35 "	786	115	0.139	"
36 "	787	120	0.133	"
37 "	788	130	0.123	"
38 "	789	150	0.106	"
39 "	790	180	0.089	"
40 "	791	200	0.080	440
41 "	792	220	0.073	440
42 "	793	240	0.067	440
43 "	794	270	0.059	580
44 "	795	300	0.053	580

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - ME
Navy - MC, AS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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